

D 120210

(Pages : 2)

Name.....

Reg. No.....

**SIXTH SEMESTER (CBCSS—UG) DEGREE EXAMINATION
MARCH 2025**

Physics/Applied Physics

PHY6B14 (EL3)—MATERIALS SCIENCE

(Admissions Year—2019 Onwards)

Time : Two Hours

Maximum : 60 Marks

*The symbols used in this question paper have their usual meanings.***Section A (Short Answer Type)**

*(Answer **all** questions in two **or** three sentences, each correct answer carries a maximum of 2 marks).*

1. What are composites ? Give the classification of composites.
2. Write a short note on thermosetting and thermoplastic polymers.
3. List the important mechanical properties of materials.
4. What is polymerization ? What are the different strengthening mechanisms of polymers ?
5. What is a semiconducting material ? Give the significance.
6. Explain Grain and Grain boundary.
7. Write short notes on Fullerenes.
8. Explain the principle of electron microscopy.
9. What are refractories ?
10. Explain Metallic bonding in crystals.
11. What are Allotropes ? Give one Example
12. What is diffusion in solids and mention the two types of diffusion in solids ?

(Ceiling = 20 marks)

Turn over

Section B (Paragraph / Problem Type)

(Answer **all** questions in a paragraph of about **half a page to one page**, each correct answer carries a maximum of 5 marks)

13. Explain the construction and working Scanning probe microscopy
14. How will you determine grain size of a sample ?
15. Find out the packing fraction of FCC.
16. Explain viscoelastic deformation of polymers.
17. Compute the line energy of dislocations in BCC iron. The Burgers vector in iron is $\frac{1}{2}$ for $\langle 111 \rangle$ type. The shear modulus of iron is 80.2 GN m^{-2} .
18. Explain the factors effecting diffusion ?
19. Explain Frenkel defects and derive an expression for the equilibrium concentration for Frenkel defects.

(Ceiling = 30 marks)

Section C (Essay Type)

(Essays - Answer in about **two pages**, any **one** question. The question carries 10 marks.

20. Explain dislocations in solids and their importance in material science.
21. Explain the principle and working of Scanning Electron Microscope.

(1 × 10 = 10 marks)